```
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
import seaborn as sns
%matplotlib inline

# Load Data
df = pd.read_csv("Retail.csv")
df

df.head()

df.describe()

df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30000 entries, 0 to 29999
Data columns (total 9 columns):
 #   Column            Non-Null Count  Dtype  
---  ------            --------------  -----  
 0   YEAR              30000 non-null  int64  
 1   MONTH             30000 non-null  int64  
 2   SUPPLIER          29967 non-null  object 
 3   ITEM CODE         30000 non-null  object 
 4   ITEM DESCRIPTION  30000 non-null  object 
 5   ITEM TYPE         30000 non-null  object 
 6   RETAIL SALES      29999 non-null  float64
 7   RETAIL TRANSFERS  30000 non-null  float64
 8   WAREHOUSE SALES   30000 non-null  float64
dtypes: float64(3), int64(2), object(4)
memory usage: 2.1+ MB
```

```
df.isnull().sum()
```

```
YEAR                 0
MONTH                0
SUPPLIER            33
ITEM CODE            0
ITEM DESCRIPTION     0
ITEM TYPE            0
RETAIL SALES         1
RETAIL TRANSFERS     0
WAREHOUSE SALES      0
dtype: int64
```

```
# Year unique Values
unique_years = df["YEAR"].unique()
unique_years
```

```
array([2020], dtype=int64)
```

```
# Month unique Values
unique_years = df["MONTH"].unique()
unique_years
```

```
array([1, 7, 3, 9], dtype=int64)
```

```
df.shape
```

```
(30000, 9)
```

```
# Sales month -> [1,3,7,9]
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30000 entries, 0 to 29999
Data columns (total 9 columns):
 #   Column            Non-Null Count  Dtype  
---  ------            --------------  -----  
 0   YEAR              30000 non-null  int64  
 1   MONTH             30000 non-null  int64  
 2   SUPPLIER          29967 non-null  object 
 3   ITEM CODE         30000 non-null  object 
 4   ITEM DESCRIPTION  30000 non-null  object 
 5   ITEM TYPE         30000 non-null  object 
 6   RETAIL SALES      29999 non-null  float64
 7   RETAIL TRANSFERS  30000 non-null  float64
 8   WAREHOUSE SALES   30000 non-null  float64
dtypes: float64(3), int64(2), object(4)
memory usage: 2.1+ MB
```

```
len(df["SUPPLIER"].unique())
```

```
291
```

```
df["SUPPLIER"].notnull()
```

```
0        True
1        True
2        True
3        True
4        True
         ...
29995    True
29996    True
29997    True
29998    True
29999    True
Name: SUPPLIER, Length: 30000, dtype: bool
```

```
df[~df["SUPPLIER"].notnull()]
```

	YEAR	MONTH	SUPPLIER	ITEM CODE	ITEM DESCRIPTION	ITEM TYPE	RETAIL SALES	RETAIL TRANSFERS	WAREHOUSE SALES
106	2020	1	NaN	107	JIGGER MEASURE SHOT GLASS	STR_SUPPLIES	14.69	18.0	0.0
188	2020	1	NaN	113	BARTENDERS BLACK BOOK	STR_SUPPLIES	0.40	0.0	0.0
231	2020	1	NaN	115	PLASTIC SHOT GLASS PACK	STR_SUPPLIES	5.71	6.0	0.0
252	2020	1	NaN	117	WHISKEY TASTING JOURNAL	STR_SUPPLIES	0.08	0.0	0.0
261	2020	1	NaN	118	PLASTIC WINE GLASS PACK	STR_SUPPLIES	7.40	10.0	0.0
281	2020	1	NaN	119	BRUMATE WINE TUMBLER	STR_SUPPLIES	9.00	32.0	0.0
295	2020	1	NaN	120	WHISKEY BARREL	STR_SUPPLIES	4.00	0.0	0.0
344	2020	1	NaN	123	HOLIDAY TOTE FOIL BAG	STR_SUPPLIES	3.89	0.0	0.0
405	2020	1	NaN	1279	EMPTY WINE KEG - KEGS	DUNNAGE	0.00	0.0	-12.0
981	2020	1	NaN	2	ICE	NON-ALCOHOL	542.00	0.0	0.0
11985	2020	1	NaN	BC	BEER CREDIT	REF	0.00	0.0	-1028.0
11986	2020	1	NaN	WC	WINE CREDIT	REF	0.00	0.0	-77.0
12118	2020	7	NaN	107	JIGGER MEASURE SHOT GLASS				
12199	2020	7	NaN	113	BARTENDERS BLACK BOOK				
12229	2020	7	NaN	115	PLASTIC SHOT GLASS PACK				
12239	2020	3	NaN	115	PLASTIC SHOT GLASS PACK				
12255	2020	7	NaN	118	PLASTIC WINE GLASS PACK				
12275	2020	7	NaN	119	BRUMATE WINE TUMBLER				
12325	2020	7	NaN	123	HOLIDAY TOTE FOIL BAG				
12382	2020	7	NaN	1279	EMPTY WINE KEG - KEGS				
12941	2020	7	NaN	2	ICE				
18390	2020	7	NaN	4	RMS ITEM				
23212	2020	7	NaN	BC	BEER CREDIT				
23213	2020	7	NaN	WC	WINE CREDIT				
23257	2020	3	NaN	123	HOLIDAY TOTE FOIL BAG				
23280	2020	3	NaN	2	ICE				
23708	2020	9	NaN	123	HOLIDAY TOTE FOIL BAG				
23825	2020	9	NaN	117	WHISKEY TASTING JOURNAL				
24438	2020	3	NaN	107	JIGGER MEASURE SHOT GLASS				
24521	2020	3	NaN	113	BARTENDERS BLACK BOOK				
24588	2020	3	NaN	118	PLASTIC WINE GLASS PACK				
24609	2020	3	NaN	119	BRUMATE WINE TUMBLER				
24719	2020	3	NaN	1279	EMPTY WINE KEG - KEGS				

```
12118  STR_SUPPLIES         12.51              17.0               0.0
12199  STR_SUPPLIES          0.20               0.0               0.0
12229  STR_SUPPLIES          5.47               8.0               0.0
12239  STR_SUPPLIES          4.74               4.0               0.0
12255  STR_SUPPLIES          7.10              14.0               0.0
12275  STR_SUPPLIES          8.00               0.0               0.0
12325  STR_SUPPLIES          0.08               0.0               0.0
12382       DUNNAGE          0.00               0.0              -2.0
12941   NON-ALCOHOL       2739.00               0.0               0.0
18390   NON-ALCOHOL           NaN               0.0               0.0
23212           REF          0.00               0.0            -190.0
23213           REF          0.00               0.0             -34.0
23257  STR_SUPPLIES          0.16               0.0               0.0
23280   NON-ALCOHOL        561.00               0.0               0.0
23708  STR_SUPPLIES          0.25               0.0               0.0
23825  STR_SUPPLIES          0.04               0.0               0.0
24438  STR_SUPPLIES         17.06              11.0               0.0
24521  STR_SUPPLIES          0.60               0.0               0.0
24588  STR_SUPPLIES          5.10               5.0               0.0
24609  STR_SUPPLIES         15.00               0.0               0.0
24719       DUNNAGE          0.00               0.0              -2.0
```

```
np.array(df["SUPPLIER"].mode())
```

```
array(['THE COUNTRY VINTNER, LLC DBA WINEBOW'], dtype=object)
```

```
df["SUPPLIER"].fillna(value="THE COUNTRY VINTNER",inplace=True)
```

```
C:\Users\v.rajkumar\AppData\Local\Temp\
ipykernel_16300\1689455609.py:1: FutureWarning: A value is trying to
be set on a copy of a DataFrame or Series through chained assignment
using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never
work because the intermediate object on which we are setting values
always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try
using 'df.method({col: value}, inplace=True)' or df[col] =
df[col].method(value) instead, to perform the operation inplace on the
original object.


  df["SUPPLIER"].fillna(value="THE COUNTRY VINTNER",inplace=True)
```

```
df.isnull().sum()
```

```
YEAR                0
MONTH               0
SUPPLIER            0
ITEM CODE           0
ITEM DESCRIPTION    0
```

```
ITEM TYPE           0
RETAIL SALES        1
RETAIL TRANSFERS    0
WAREHOUSE SALES     0
dtype: int64
```

```
df[~df["SUPPLIER"].notnull()]
```

```
Empty DataFrame
Columns: [YEAR, MONTH, SUPPLIER, ITEM CODE, ITEM DESCRIPTION, ITEM
TYPE, RETAIL SALES, RETAIL TRANSFERS, WAREHOUSE SALES]
Index: []
```

```
df["ITEM CODE"]
```

```
0        100009
1        100024
2          1001
3        100145
4        100293
          ...
29995    352322
29996    352324
29997    352354
29998     35238
29999    352380
Name: ITEM CODE, Length: 30000, dtype: object
```

```
df["ITEM DESCRIPTION"].head()
```

```
0                    BOOTLEG RED - 750ML
1              MOMENT DE PLAISIR - 750ML
2    S SMITH ORGANIC PEAR CIDER - 18.7OZ
3          SCHLINK HAUS KABINETT - 750ML
4         SANTORINI GAVALA WHITE - 750ML
Name: ITEM DESCRIPTION, dtype: object
```

```
df["ITEM TYPE"]
```

```
0          WINE
1          WINE
2          BEER
3          WINE
4          WINE
          ...
29995    LIQUOR
29996      WINE
29997      WINE
29998    LIQUOR
29999      WINE
Name: ITEM TYPE, Length: 30000, dtype: object
```

```
df["ITEM TYPE"].unique()
```

```
array(['WINE', 'BEER', 'LIQUOR', 'STR_SUPPLIES', 'KEGS', 'REF',
'DUNNAGE',
       'NON-ALCOHOL'], dtype=object)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30000 entries, 0 to 29999
Data columns (total 9 columns):
 #   Column            Non-Null Count  Dtype  
---  ------            --------------  -----  
 0   YEAR              30000 non-null  int64  
 1   MONTH             30000 non-null  int64  
 2   SUPPLIER          30000 non-null  object 
 3   ITEM CODE         30000 non-null  object 
 4   ITEM DESCRIPTION  30000 non-null  object 
 5   ITEM TYPE         30000 non-null  object 
 6   RETAIL SALES      29999 non-null  float64
 7   RETAIL TRANSFERS  30000 non-null  float64
 8   WAREHOUSE SALES   30000 non-null  float64
dtypes: float64(3), int64(2), object(4)
memory usage: 2.1+ MB
```

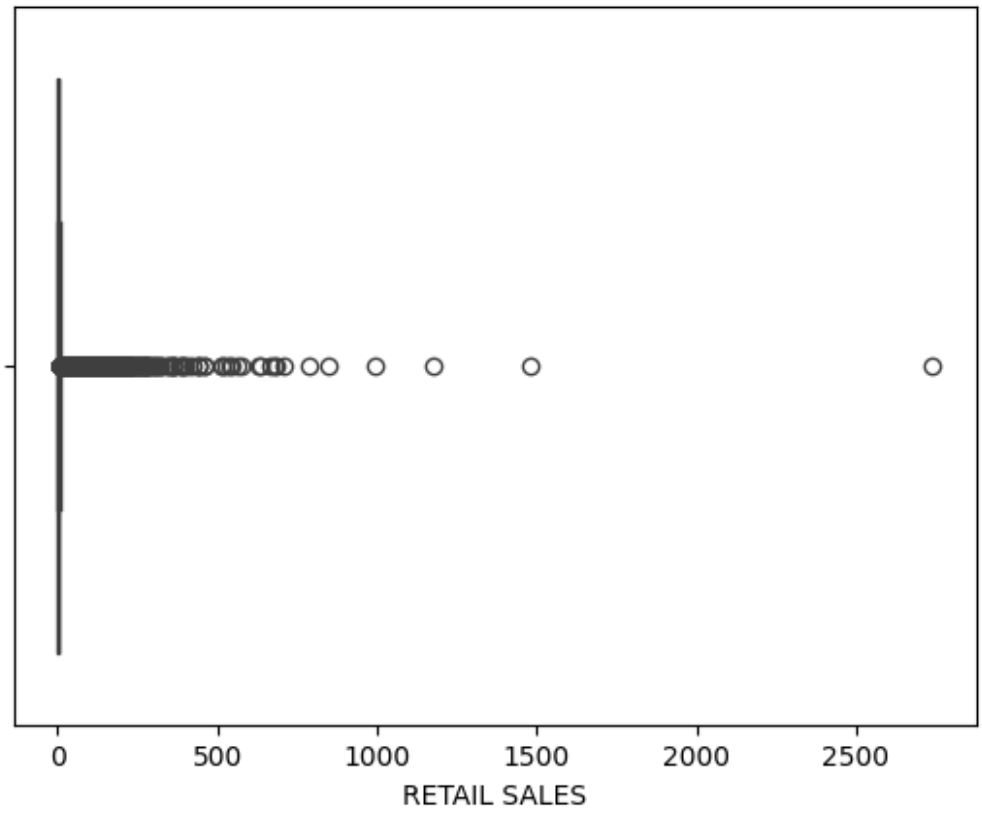
```
df[~df["RETAIL SALES"].notnull()]
```

```
       YEAR  MONTH             SUPPLIER ITEM CODE ITEM DESCRIPTION  \
18390  2020      7  THE COUNTRY VINTNER         4         RMS ITEM   

         ITEM TYPE  RETAIL SALES  RETAIL TRANSFERS  WAREHOUSE SALES
18390  NON-ALCOHOL           NaN               0.0              0.0
```

```
sns.boxplot(data=df,x="RETAIL SALES")
```

```
<Axes: xlabel='RETAIL SALES'>
```




```
median_rs = df["RETAIL SALES"].median()
median_rs
```

```
0.16
```

```
df["RETAIL SALES"].fillna(value=median_rs,inplace=True)
```

```
C:\Users\v.rajkumar\AppData\Local\Temp\
ipykernel_16300\1529794574.py:1: FutureWarning: A value is trying to
be set on a copy of a DataFrame or Series through chained assignment
using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never
work because the intermediate object on which we are setting values
always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try
using 'df.method({col: value}, inplace=True)' or df[col] =
df[col].method(value) instead, to perform the operation inplace on the
original object.


  df["RETAIL SALES"].fillna(value=median_rs,inplace=True)
```

```
df["RETAIL SALES"]
```

```
0          0.00
1          0.00
2          0.00
3          0.00
4          0.82
          ...
29995      0.33
29996      0.00
29997      0.00
29998    329.04
29999      0.00
Name: RETAIL SALES, Length: 30000, dtype: float64
```

```
df["RETAIL TRANSFERS"].isnull().sum()
```

```
0
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30000 entries, 0 to 29999
Data columns (total 9 columns):
 #   Column            Non-Null Count  Dtype  
---  ------            --------------  -----  
 0   YEAR              30000 non-null  int64  
 1   MONTH             30000 non-null  int64  
 2   SUPPLIER          30000 non-null  object 
 3   ITEM CODE         30000 non-null  object 
 4   ITEM DESCRIPTION  30000 non-null  object 
 5   ITEM TYPE         30000 non-null  object 
 6   RETAIL SALES      30000 non-null  float64
 7   RETAIL TRANSFERS  30000 non-null  float64
 8   WAREHOUSE SALES   30000 non-null  float64
dtypes: float64(3), int64(2), object(4)
memory usage: 2.1+ MB
```

```
df["WAREHOUSE SALES"]
```

```
0        2.0
1        4.0
2        1.0
3        1.0
4        0.0
        ...
29995    0.0
29996    0.0
29997    6.0
29998    0.0
29999    1.0
Name: WAREHOUSE SALES, Length: 30000, dtype: float64
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30000 entries, 0 to 29999
Data columns (total 9 columns):
 #   Column            Non-Null Count  Dtype  
---  ------            --------------  -----  
 0   YEAR              30000 non-null  int64  
 1   MONTH             30000 non-null  int64  
 2   SUPPLIER          30000 non-null  object 
 3   ITEM CODE         30000 non-null  object 
 4   ITEM DESCRIPTION  30000 non-null  object 
 5   ITEM TYPE         30000 non-null  object 
 6   RETAIL SALES      30000 non-null  float64
 7   RETAIL TRANSFERS  30000 non-null  float64
 8   WAREHOUSE SALES   30000 non-null  float64
dtypes: float64(3), int64(2), object(4)
memory usage: 2.1+ MB
```

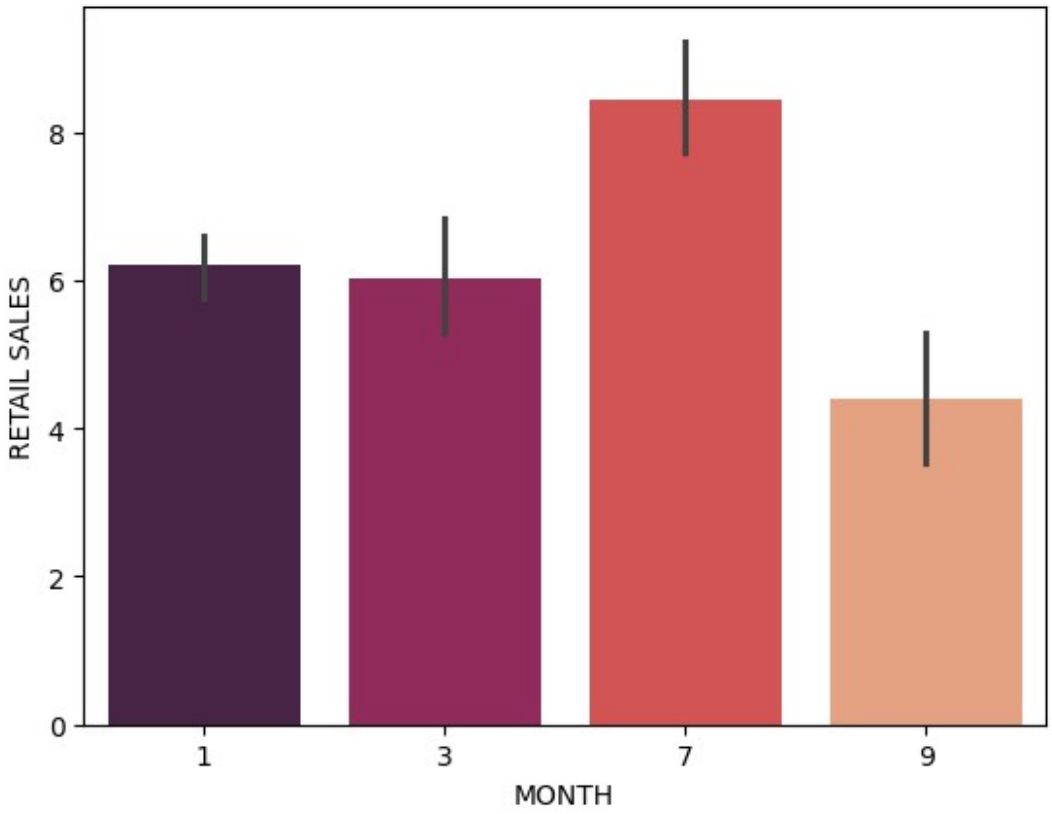
```
sns.barplot(data=df,x="MONTH",y="RETAIL SALES",palette="rocket")
```

```
C:\Users\v.rajkumar\AppData\Local\Temp\ipykernel_16300\674477137.py:1: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.barplot(data=df,x="MONTH",y="RETAIL SALES",palette="rocket")
```

```
<Axes: xlabel='MONTH', ylabel='RETAIL SALES'>
```




```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30000 entries, 0 to 29999
Data columns (total 9 columns):
 #   Column            Non-Null Count  Dtype  
---  ------            --------------  -----  
 0   YEAR              30000 non-null  int64  
 1   MONTH             30000 non-null  int64  
 2   SUPPLIER          30000 non-null  object 
 3   ITEM CODE         30000 non-null  object 
 4   ITEM DESCRIPTION  30000 non-null  object 
 5   ITEM TYPE         30000 non-null  object 
 6   RETAIL SALES      30000 non-null  float64
 7   RETAIL TRANSFERS  30000 non-null  float64
 8   WAREHOUSE SALES   30000 non-null  float64
dtypes: float64(3), int64(2), object(4)
memory usage: 2.1+ MB
```

```
max_supplier_sale = df.groupby("SUPPLIER")["RETAIL
SALES"].max().sort_values(ascending=False)
```

```
max_supplier_sale.head()
```

```
SUPPLIER
THE COUNTRY VINTNER       2739.00
FIFTH GENERATION INC      1482.17
CROWN IMPORTS             1174.00
HEINEKEN USA               787.00
MILLER BREWING COMPANY     663.00
Name: RETAIL SALES, dtype: float64
```

```
total_top_5_supplier_sale = df.groupby("SUPPLIER")["RETAIL
SALES"].sum().sort_values(ascending=False)
```

```
total_top_5_supplier_sale.head(n=5)
```

```
SUPPLIER
DIAGEO NORTH AMERICA INC    13840.54
E & J GALLO WINERY          12742.24
ANHEUSER BUSCH INC          10091.64
CONSTELLATION BRANDS         9931.78
JIM BEAM BRANDS CO           9845.79
Name: RETAIL SALES, dtype: float64
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30000 entries, 0 to 29999
Data columns (total 9 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   YEAR              30000 non-null  int64
 1   MONTH             30000 non-null  int64
 2   SUPPLIER          30000 non-null  object
 3   ITEM CODE         30000 non-null  object
 4   ITEM DESCRIPTION  30000 non-null  object
 5   ITEM TYPE         30000 non-null  object
 6   RETAIL SALES      30000 non-null  float64
 7   RETAIL TRANSFERS  30000 non-null  float64
 8   WAREHOUSE SALES   30000 non-null  float64
dtypes: float64(3), int64(2), object(4)
memory usage: 2.1+ MB
```

```
total_sale_count_by_supplier = df.groupby("SUPPLIER")["RETAIL
SALES"].count().sort_values(ascending=False)
```

```
total_sale_count_by_supplier.head()
```

```
SUPPLIER
THE COUNTRY VINTNER, LLC DBA WINEBOW    2041
REPUBLIC NATIONAL DISTRIBUTING CO       1832
SOUTHERN GLAZERS WINE AND SPIRITS       1304
LEGENDS LTD                             1227
```

```
MONSIEUR TOUTON SELECTION                1131
Name: RETAIL SALES, dtype: int64

df[df["SUPPLIER"]=="THE COUNTRY VINTNER"]

       YEAR  MONTH              SUPPLIER ITEM CODE           ITEM
DESCRIPTION  \
106    2020      1  THE COUNTRY VINTNER       107  JIGGER MEASURE SHOT
GLASS
188    2020      1  THE COUNTRY VINTNER       113     BARTENDERS BLACK
BOOK
231    2020      1  THE COUNTRY VINTNER       115    PLASTIC SHOT
GLASS PACK
252    2020      1  THE COUNTRY VINTNER       117    WHISKEY TASTING
JOURNAL
261    2020      1  THE COUNTRY VINTNER       118    PLASTIC WINE
GLASS PACK
281    2020      1  THE COUNTRY VINTNER       119       BRUMATE WINE
TUMBLER
295    2020      1  THE COUNTRY VINTNER       120            WHISKEY
BARREL
344    2020      1  THE COUNTRY VINTNER       123      HOLIDAY TOTE
FOIL BAG
405    2020      1  THE COUNTRY VINTNER      1279      EMPTY WINE KEG
- KEGS
981    2020      1  THE COUNTRY VINTNER         2
ICE
11985  2020      1  THE COUNTRY VINTNER        BC               BEER
CREDIT
11986  2020      1  THE COUNTRY VINTNER        WC               WINE
CREDIT
12118  2020      7  THE COUNTRY VINTNER       107  JIGGER MEASURE SHOT
GLASS
12199  2020      7  THE COUNTRY VINTNER       113     BARTENDERS BLACK
BOOK
12229  2020      7  THE COUNTRY VINTNER       115    PLASTIC SHOT
GLASS PACK
12239  2020      3  THE COUNTRY VINTNER       115    PLASTIC SHOT
GLASS PACK
12255  2020      7  THE COUNTRY VINTNER       118    PLASTIC WINE
GLASS PACK
12275  2020      7  THE COUNTRY VINTNER       119       BRUMATE WINE
TUMBLER
12325  2020      7  THE COUNTRY VINTNER       123      HOLIDAY TOTE
FOIL BAG
12382  2020      7  THE COUNTRY VINTNER      1279      EMPTY WINE KEG
- KEGS
12941  2020      7  THE COUNTRY VINTNER         2
ICE
18390  2020      7  THE COUNTRY VINTNER         4
```

```
RMS ITEM
23212  2020      7  THE COUNTRY VINTNER        BC                BEER
CREDIT
23213  2020      7  THE COUNTRY VINTNER        WC                WINE
CREDIT
23257  2020      3  THE COUNTRY VINTNER       123      HOLIDAY TOTE
FOIL BAG
23280  2020      3  THE COUNTRY VINTNER         2
ICE
23708  2020      9  THE COUNTRY VINTNER       123      HOLIDAY TOTE
FOIL BAG
23825  2020      9  THE COUNTRY VINTNER       117    WHISKEY TASTING
JOURNAL
24438  2020      3  THE COUNTRY VINTNER       107  JIGGER MEASURE SHOT
GLASS
24521  2020      3  THE COUNTRY VINTNER       113     BARTENDERS BLACK
BOOK
24588  2020      3  THE COUNTRY VINTNER       118    PLASTIC WINE
GLASS PACK
24609  2020      3  THE COUNTRY VINTNER       119       BRUMATE WINE
TUMBLER
24719  2020      3  THE COUNTRY VINTNER      1279      EMPTY WINE KEG
- KEGS

          ITEM TYPE  RETAIL SALES  RETAIL TRANSFERS  WAREHOUSE SALES
106    STR_SUPPLIES         14.69              18.0              0.0
188    STR_SUPPLIES          0.40               0.0              0.0
231    STR_SUPPLIES          5.71               6.0              0.0
252    STR_SUPPLIES          0.08               0.0              0.0
261    STR_SUPPLIES          7.40              10.0              0.0
281    STR_SUPPLIES          9.00              32.0              0.0
295    STR_SUPPLIES          4.00               0.0              0.0
344    STR_SUPPLIES          3.89               0.0              0.0
405         DUNNAGE          0.00               0.0            -12.0
981     NON-ALCOHOL        542.00               0.0              0.0
11985           REF          0.00               0.0          -1028.0
11986           REF          0.00               0.0            -77.0
12118  STR_SUPPLIES         12.51              17.0              0.0
12199  STR_SUPPLIES          0.20               0.0              0.0
12229  STR_SUPPLIES          5.47               8.0              0.0
12239  STR_SUPPLIES          4.74               4.0              0.0
12255  STR_SUPPLIES          7.10              14.0              0.0
12275  STR_SUPPLIES          8.00               0.0              0.0
12325  STR_SUPPLIES          0.08               0.0              0.0
12382       DUNNAGE          0.00               0.0             -2.0
12941   NON-ALCOHOL       2739.00               0.0              0.0
18390   NON-ALCOHOL          0.16               0.0              0.0
23212           REF          0.00               0.0           -190.0
23213           REF          0.00               0.0            -34.0
```

```
23257  STR_SUPPLIES          0.16               0.0               0.0
23280   NON-ALCOHOL        561.00               0.0               0.0
23708  STR_SUPPLIES          0.25               0.0               0.0
23825  STR_SUPPLIES          0.04               0.0               0.0
24438  STR_SUPPLIES         17.06              11.0               0.0
24521  STR_SUPPLIES          0.60               0.0               0.0
24588  STR_SUPPLIES          5.10               5.0               0.0
24609  STR_SUPPLIES         15.00               0.0               0.0
24719       DUNNAGE          0.00               0.0              -2.0
```

```
df.duplicated().sum()
```

```
0
```

```
df_corr = df.select_dtypes(include=["float","int"])
```

```
df_corr
```

```
       YEAR  MONTH  RETAIL SALES  RETAIL TRANSFERS  WAREHOUSE SALES
0      2020      1          0.00               0.0              2.0
1      2020      1          0.00               1.0              4.0
2      2020      1          0.00               0.0              1.0
3      2020      1          0.00               0.0              1.0
4      2020      1          0.82               0.0              0.0
...     ...    ...           ...               ...              ...
29995  2020      3          0.33               0.0              0.0
29996  2020      3          0.00               0.0              0.0
29997  2020      3          0.00               0.0              6.0
29998  2020      3        329.04             302.0              0.0
29999  2020      3          0.00               0.0              1.0

[30000 rows x 5 columns]
```

```
df_corr.corr()
```

```
                   YEAR     MONTH  RETAIL SALES  RETAIL TRANSFERS  \
YEAR                NaN       NaN           NaN               NaN
MONTH               NaN  1.000000      0.024366          0.011043
RETAIL SALES        NaN  0.024366      1.000000          0.844598
RETAIL TRANSFERS    NaN  0.011043      0.844598          1.000000
WAREHOUSE SALES     NaN  0.017208      0.491823          0.540071

                   WAREHOUSE SALES
YEAR                           NaN
MONTH                     0.017208
RETAIL SALES              0.491823
RETAIL TRANSFERS          0.540071
WAREHOUSE SALES           1.000000
```

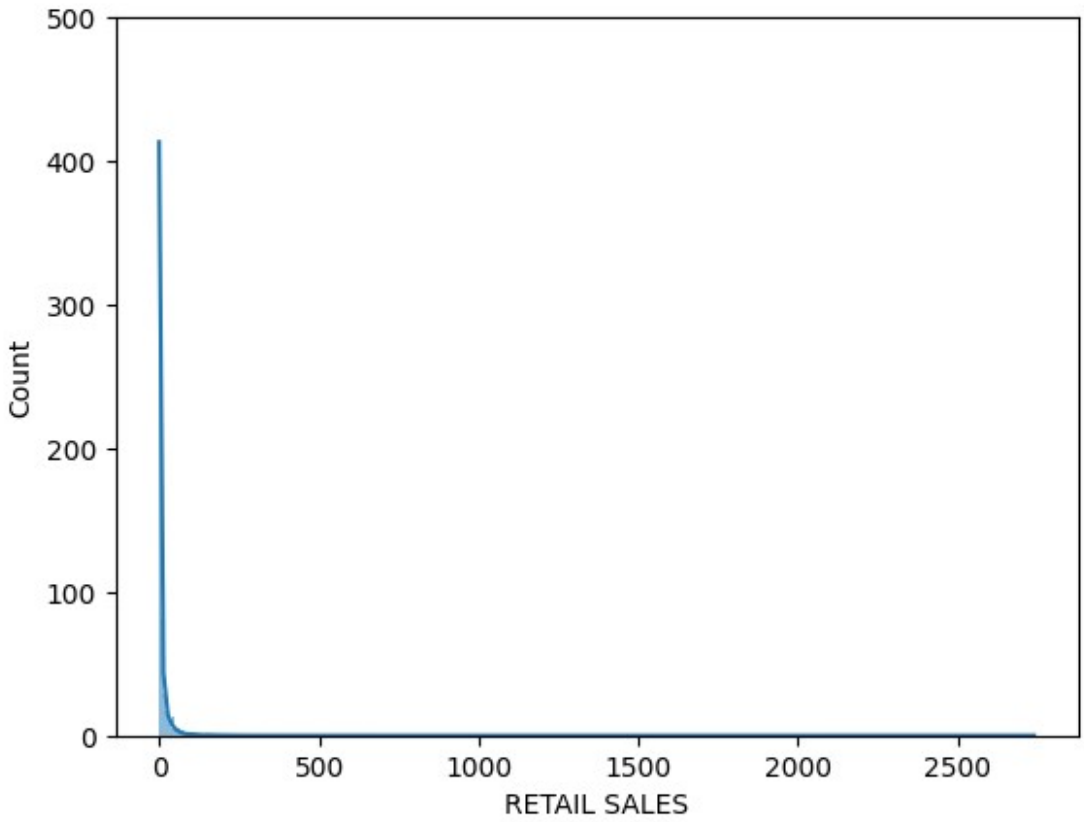
```
categorical_analysis = df.groupby("SUPPLIER")["ITEM TYPE"].max()
```

```
categorical_analysis
```

```
SUPPLIER
A I G WINE & SPIRITS                  WINE
ADAMBA IMPORTS INTL                 LIQUOR
AIKO IMPORTERS INC                    WINE
ALLAGASH BREWING COMPANY              KEGS
ALLIED IMPORTERS USA LTD              WINE
                                       ...
WINERY EXCHANGE INC                   WINE
YOUNG WON TRADING INC                 WINE
YUENGLING BREWERY                     KEGS
Z WINE GALLERY IMPORTS LLC            WINE
ZURENA LLC                       NON-ALCOHOL
Name: ITEM TYPE, Length: 291, dtype: object
```

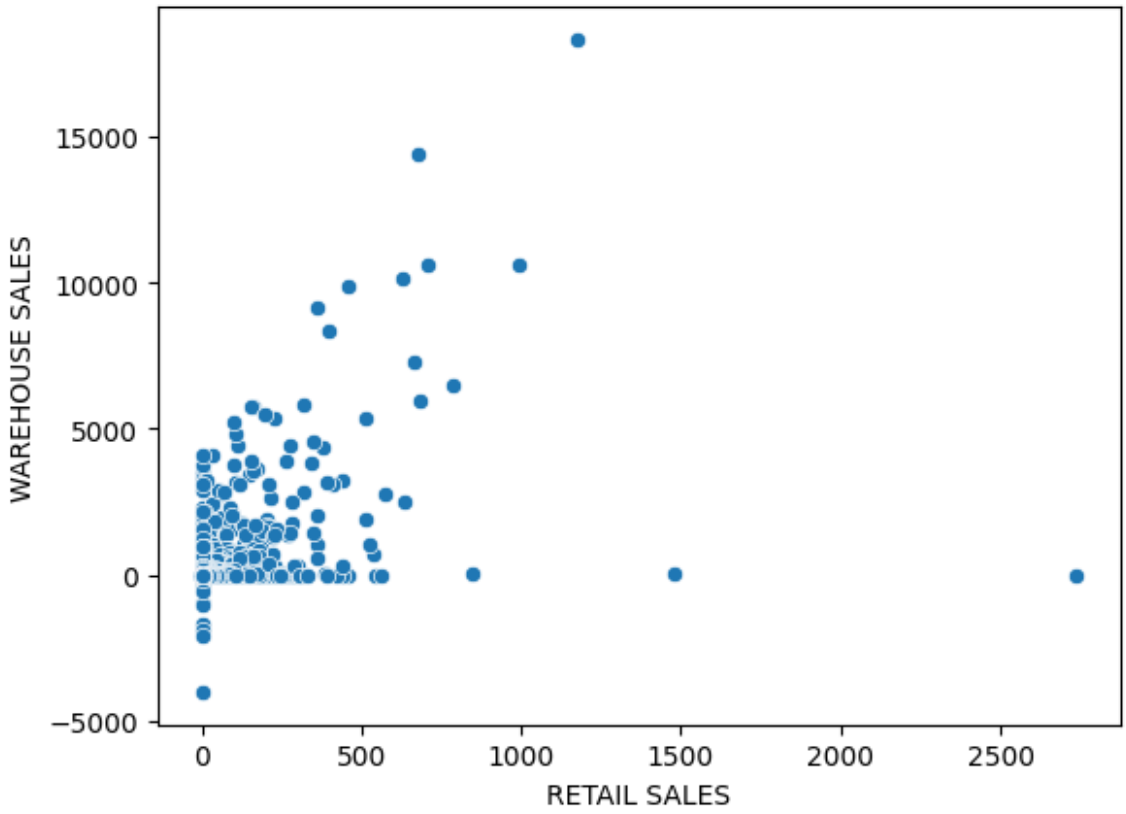
```
sns.histplot(data=df,x="RETAIL SALES",kde=True)
plt.ylim(0,500)
```

```
(0.0, 500.0)
```



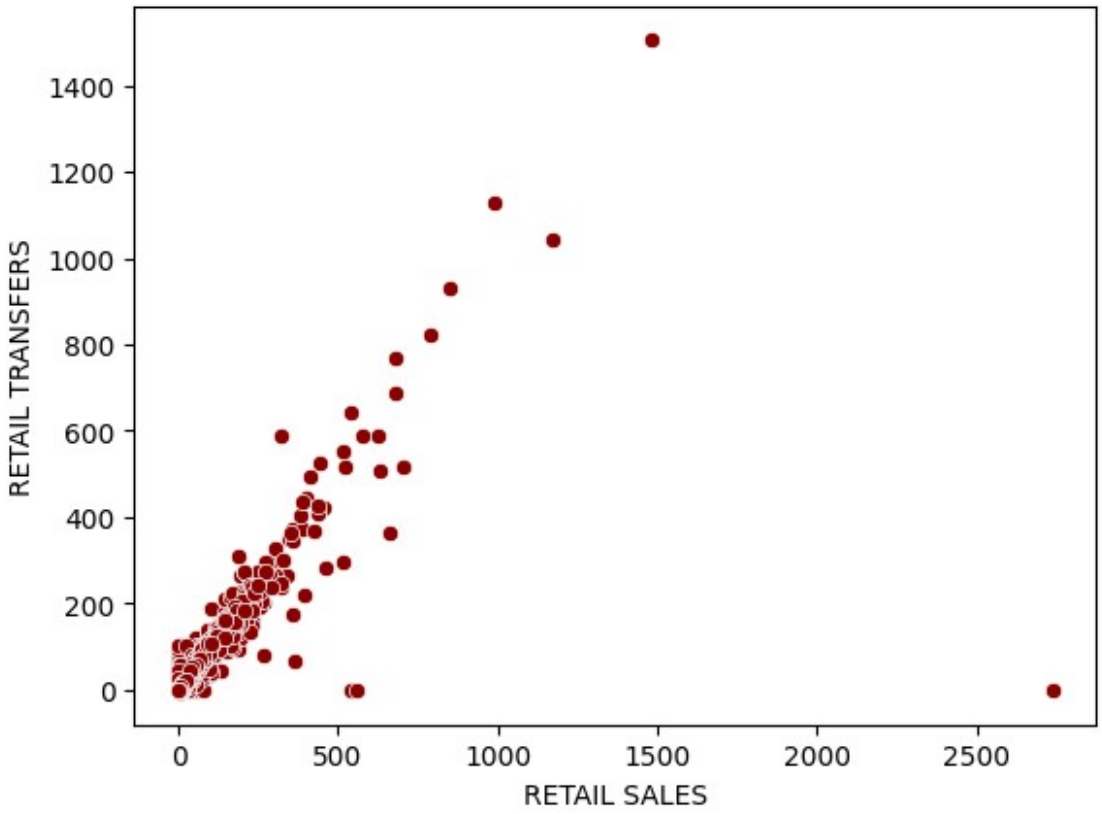

```
sns.scatterplot(data=df,x="RETAIL SALES",y="WAREHOUSE SALES")
```

```
<Axes: xlabel='RETAIL SALES', ylabel='WAREHOUSE SALES'>
```



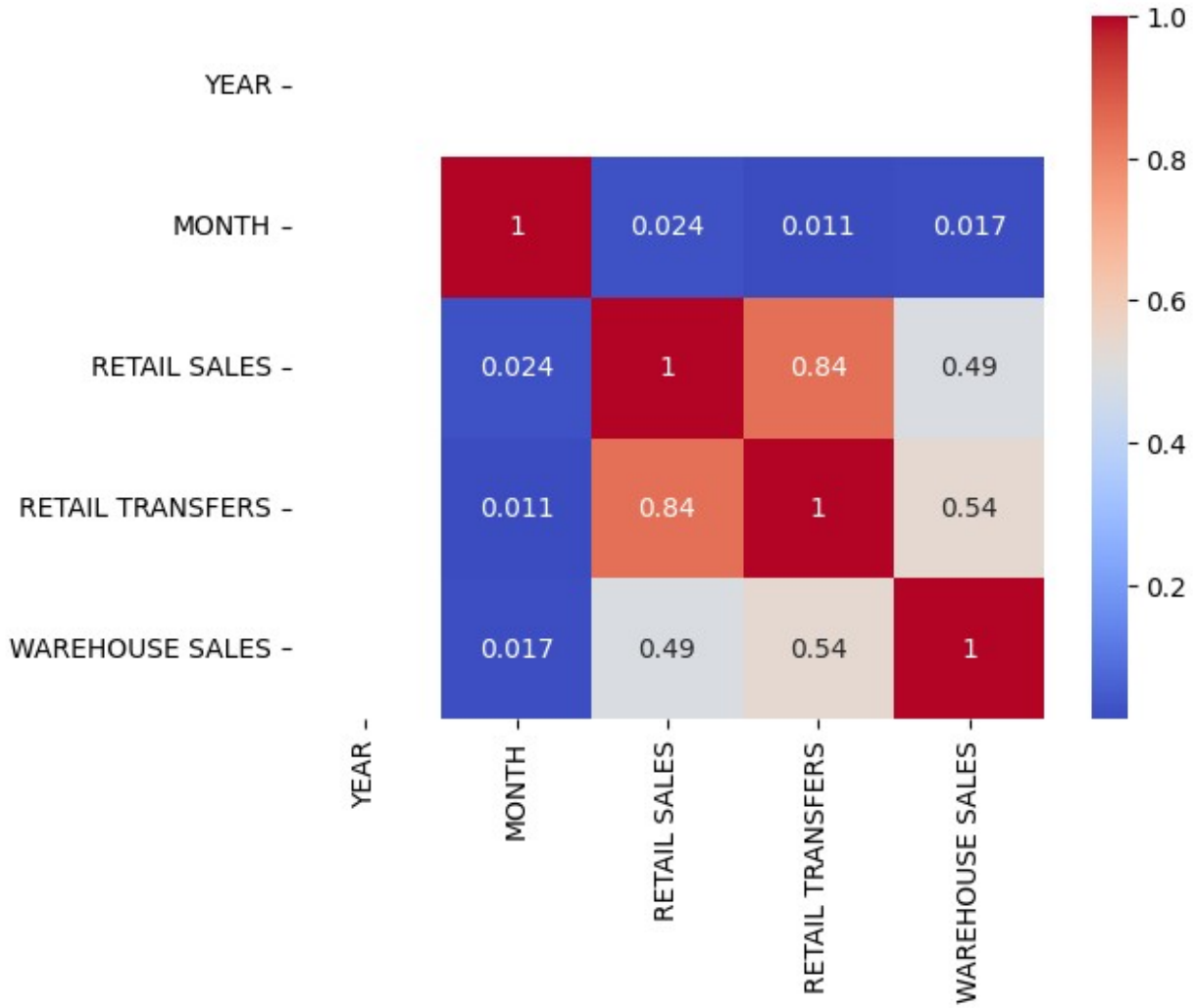

```
sns.scatterplot(data=df,x="RETAIL SALES",y="RETAIL
TRANSFERS",color="darkred")

<Axes: xlabel='RETAIL SALES', ylabel='RETAIL TRANSFERS'>
```




```
sns.heatmap(df_corr.corr(),annot=True,cmap="coolwarm")
```

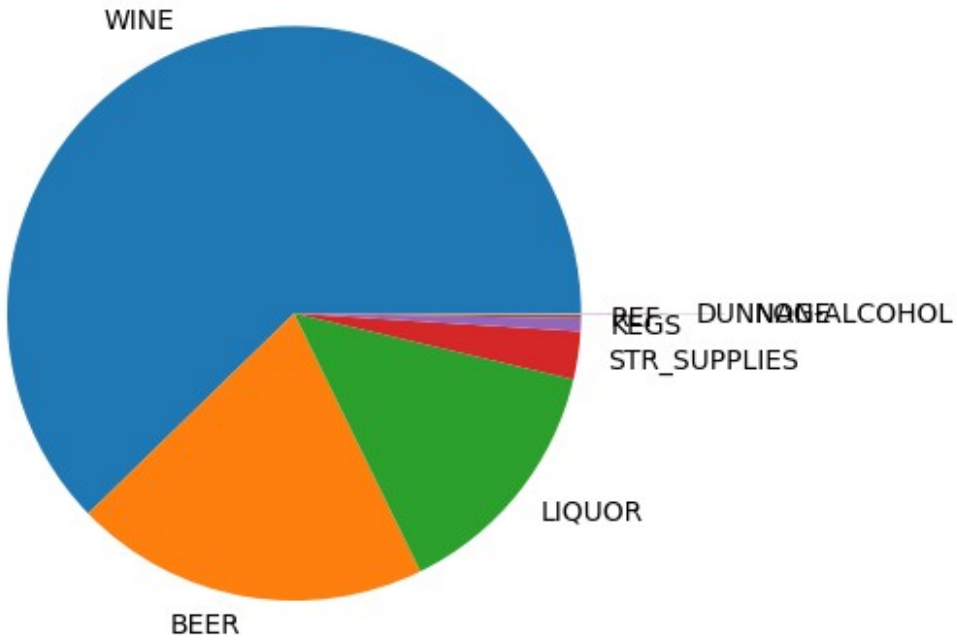
```
<Axes: >
```




```
plt.pie(df["ITEM TYPE"].value_counts(),labels=df["ITEM
TYPE"].unique(),explode=[0,0,0,0,0,0,0.3,0.5])
```

```
([<matplotlib.patches.Wedge at 0x1cbff3b1db0>,
  <matplotlib.patches.Wedge at 0x1cbff3b1cf0>,
  <matplotlib.patches.Wedge at 0x1cbff3b2680>,
  <matplotlib.patches.Wedge at 0x1cbff3b2b00>,
  <matplotlib.patches.Wedge at 0x1cbff3b2f80>,
  <matplotlib.patches.Wedge at 0x1cbff3b3400>,
  <matplotlib.patches.Wedge at 0x1cbff3b3880>,
  <matplotlib.patches.Wedge at 0x1cbff3b3d00>],
 [Text(-0.4134908717440754, 1.0193259042054827, 'WINE'),
  Text(-0.18855466764695003, -1.0837191228858833, 'BEER'),
  Text(0.8596964231135443, -0.6862376119725426, 'LIQUOR'),
  Text(1.0880645492638197, -0.16160302173945032, 'STR_SUPPLIES'),
  Text(1.099110745951579, -0.044221806088891494, 'KEGS'),
  Text(1.0999335030826436, -0.012094990547504866, 'REF'),
```

```
  Text(1.3999935437896722, -0.004251746374808932, 'DUNNAGE'),
  Text(1.599999438403932, -0.001340562233817801, 'NON-ALCOHOL')])
```




```
from statsmodels.tsa.seasonal import seasonal_decompose
from statsmodels.tsa.arima.model import ARIMA

decompose_result = seasonal_decompose(df["RETAIL
SALES"],model="additive",period=12)
arima = ARIMA(df["RETAIL SALES"],order=(1,1,1))
arima_fit = arima.fit()
arima_fit.summary()

<class 'statsmodels.iolib.summary.Summary'>
"""
                               SARIMAX Results

==========================================================================
========
Dep. Variable:           RETAIL SALES   No. Observations:
30000
Model:                 ARIMA(1, 1, 1)   Log Likelihood              -
147310.754
Date:                Fri, 03 Oct 2025   AIC
294627.508
Time:                        08:42:50   BIC
294652.435
```

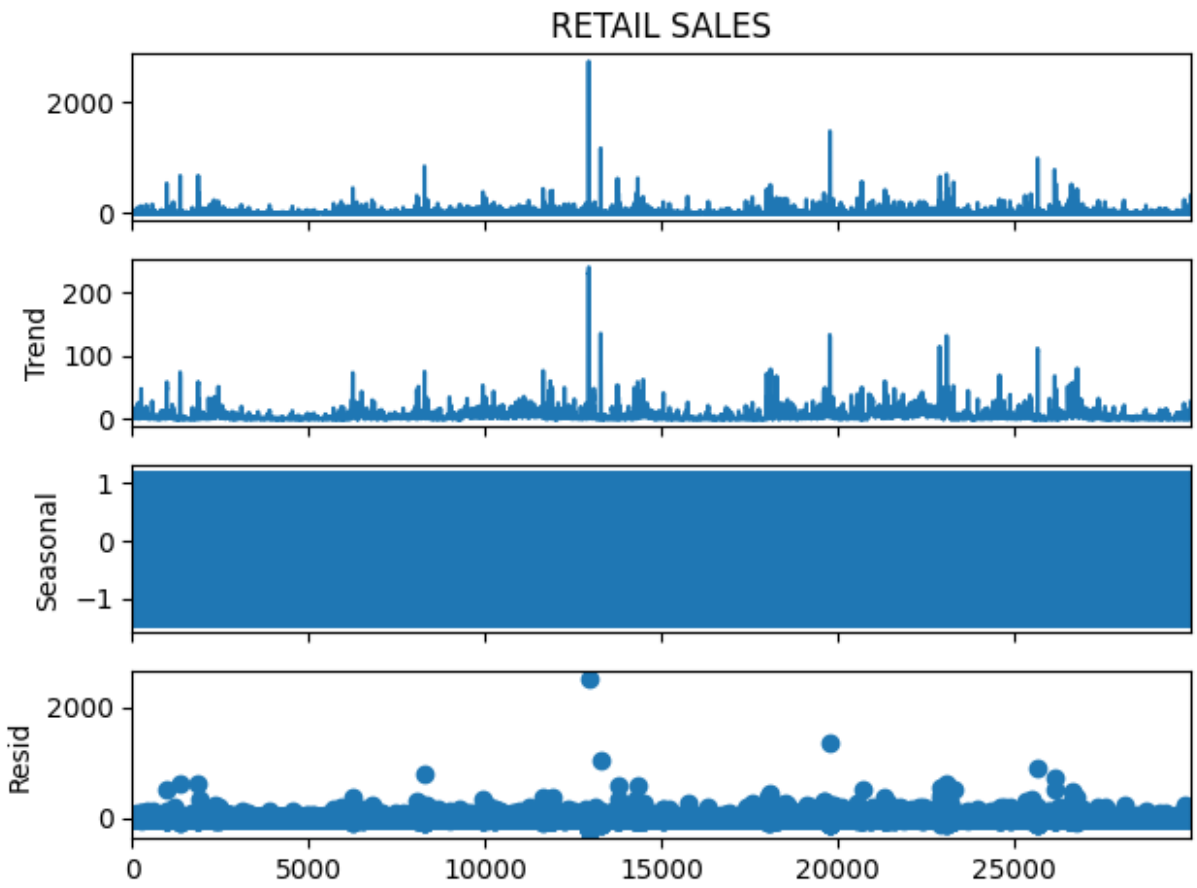
```
Sample:                                0   HQIC
294635.506
                               - 30000

Covariance Type:                   opg

==============================================================================
========
                 coef    std err          z      P>|z|      [0.025
0.975]
------------------------------------------------------------------------------
--------
ar.L1          0.0288      0.003     10.139      0.000       0.023
0.034
ma.L1         -0.9905      0.001  -1445.550      0.000      -0.992
-0.989
sigma2      1078.1936      0.286   3766.374      0.000    1077.633
1078.755
==============================================================================
==============
Ljung-Box (L1) (Q):                   0.00   Jarque-Bera (JB):
4553299529.10
Prob(Q):                              0.97   Prob(JB):
0.00
Heteroskedasticity (H):               1.95   Skew:
30.70
Prob(H) (two-sided):                  0.00   Kurtosis:
1910.61
==============================================================================
==============

Warnings:
[1] Covariance matrix calculated using the outer product of gradients
(complex-step).
"""

decompose_result.plot()
```




```
model_forecast = arima_fit.forecast(steps=12)
model_forecast
```

```
30000    6.201007
30001    6.379894
30002    6.385055
30003    6.385204
30004    6.385208
30005    6.385208
30006    6.385208
30007    6.385208
30008    6.385208
30009    6.385208
30010    6.385208
30011    6.385208
Name: predicted_mean, dtype: float64
```

```
df.to_csv("cleaned_retail.csv",index=False)
```

Conclusion

- Sales - retail shop, retail transfers,warehouse.
- Month - It takes part in the Month of 1,3,7,9.
- Supplier - Around 292 suppliers - ['THE COUNTRY VINTNER, LLC DBA WINEBOW'] secured more number of Sales [2041 of 30000] in retail sale - total Top 2 sale supplier [DIAGEO NORTH AMERICA INC,E & J GALLO WINERY]
- Correlation - Retail Sales and retail transfers have positive correlation,Retail Sales and Warehouse have no correlation.
- Item type - Wine and Beer are got more selled item.